Experiment No . 9

AIM :

To implement client-side storage using localStorage and sessionStorage in JavaScript to save and manage user preferences such as theme settings.

SOFTWARE / TOOLS REQUIRED :

- Visual Studio Code (VS Code) or any text/code editor
- Google Chrome (or any modern web browser) to render HTML and run JavaScript
- Basic knowledge of HTML5, CSS and JavaScript

THEORY :

1. **HTML5 Web Storage:**
   Web Storage APIs allow web applications to store key-value data in the user's browser without requiring a server-side database. Data is stored as strings and is accessed using simple methods of the window.localStorage and window.sessionStorage objects.
2. **localStorage and sessionStorage:**
   `localStorage` stores data with no expiry, so the saved preference remains available even after the browser is closed and reopened. `sessionStorage` stores data only for the current browser session (tab), and it is cleared when the session ends.
3. **Dynamic Theme Switching:**
   JavaScript reads the user's selected theme (Light or Dark), saves it in the chosen storage, and changes the page appearance dynamically. When the page loads again, the stored value is retrieved and the matching theme is restored.

A. Web Storage Methods

Method	Type / Return	Description
`setItem(key, value)`	Stores (Returns undefined)	Saves a key-value pair in the storage.
`getItem(key)`	Retrieves (Returns value or null)	Reads the value stored against the given key.
`removeItem(key)`	Mutates (Returns undefined)	Removes the stored item for the given key.
`clear()`	Mutates (Returns undefined)	Removes all items from the storage.

B. localStorage vs sessionStorage

Feature	localStorage	sessionStorage
Lifetime	Persists until cleared manually	Lasts only for the current session
Scope	Shared across tabs of the same origin	Limited to a single tab/window
Capacity	About 5 MB per origin	About 5 MB per origin
Sent to server	No	No

CASE STUDY TITLE :

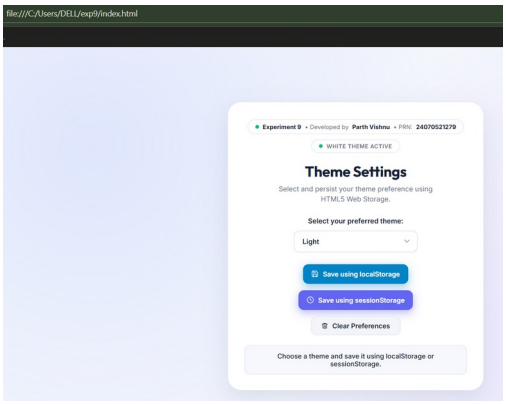
Theme Settings Web Application using localStorage and sessionStorage

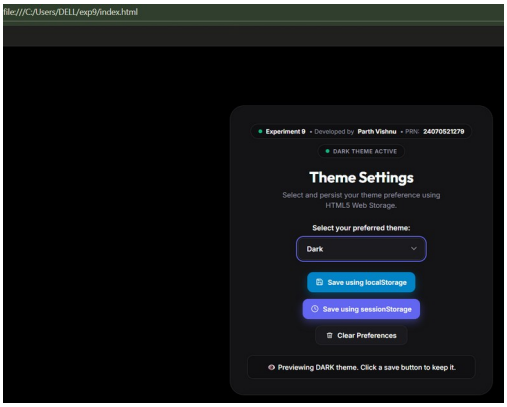
AIM OF THE CASE STUDY :

Develop a Theme Settings webpage in which the user selects a Light or Dark theme and saves the preference using either `localStorage` or `sessionStorage`. The saved theme must be restored automatically when the page is loaded again, and the user must be able to clear the stored preferences.

Name : Parth Vishnu PRN : 24070521279

File path : file:///C:/Users/DELL/exp9/index.html





OUTPUT :

On opening the page and selecting a theme, the program yields:

- **Light theme:** the page opens in the white interface, the status shows "WHITE THEME ACTIVE", and the message box prompts the user to save a theme.
- **Dark theme:** selecting Dark switches the page to the dark interface instantly, the status changes to "DARK THEME ACTIVE", and a preview message is shown.

CONCLUSION :

By completing this experiment, we have successfully studied and implemented HTML5 Web Storage using `localStorage` and `sessionStorage` through a functional Theme Settings application.

The application stored the user's theme preference (`setItem`), retrieved previously saved preferences (`getItem`), removed them when required (`removeItem`), dynamically switched between light and dark themes, and provided appropriate feedback to the user. Thus, the experiment demonstrated effective use of client-side storage and JavaScript for maintaining user preferences in a web application.